

FIN Research Monograph — Release 10.6

FIN Programs 51–60

Green Response, Fractal Information Compression,
and the Minimal Bridge to Physics

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Scope. Ten executed programs testing Green response, corrected information geometry, support transfer, exact fractal Schur compression, spectral dimension, chiral response, operational records, and the dimensional bridge to physics.

Ontology and guardrail. The nadsoliton is treated as primordial information in a solitonic state. No lower informational layer, silent legacy–strict identity, role transfer, strict selector, physical unit, or Theory-of-Everything closure is claimed.

Reproducibility. `fin_programs_51_60_green_fractal_physics.py`, `test_fin_programs_51_60_green_fractal_physics.py`, JSON, and figures.

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Contents

0.1	Confidence convention	1
1	Executive summary	2
2	Audit of the two source notes	4
2.1	What survives	4
2.2	Required corrections	4
2.3	Corrected conceptual position	5
3	Minimal mathematical setup	6
4	Program 51 — Resolvent-domain and positivity audit	7
4.1	Question	7
4.2	Result	7
4.3	Verdict	7
5	Program 52 — Corrected Green geometry	8
5.1	Question	8
5.2	Theorem	8
5.3	Explicit counterexample	8
5.4	FIN finite results	9
5.5	Verdict	9
6	Program 53 — Resolvent reconstruction of dual dynamics	10
6.1	Question	10
6.2	Test	10
6.3	Results	10
6.4	Verdict	10
7	Program 54 — Functional Green bridge and transfer falsification	12
7.1	Question	12
7.2	Results	12

7.3	Interpretation	12
7.4	Verdict	13
8	Program 55 — Green-Schur fractal compression	14
8.1	Question	14
8.2	New typed compression object	14
8.3	Results	14
8.4	Verdict	15
9	Program 56 — Spectral dimension under repair and scale	16
9.1	Question	16
9.2	Results	16
9.3	Verdict	16
10	Program 57 — Information action and compression loss	18
10.1	Question	18
10.2	Results	18
10.3	Verdict	18
11	Program 58 — Chiral Green response under fractal compression	20
11.1	Question	20
11.2	Results	20
11.3	Verdict	20
12	Program 59 — Operational records from one generator	21
12.1	Question	21
12.2	Results	21
12.3	Verdict	21
13	Program 60 — Physical-scale obstruction and minimal conditioned bridge	23
13.1	Question	23
13.2	Impossibility theorem	23
13.3	Minimal conditioned bridge	24
13.3.1	W_0 : informational core	24
13.3.2	CA: conversion axioms	24
13.3.3	OA: operational axioms	24
13.3.4	SA: sector axioms, only when required	24
13.3.5	W_3 : conditioned physics	24
13.4	Verdict	25

14 Integrated scientific conclusions	26
14.1 What the Green function genuinely unifies	26
14.2 What it does not unify	26
14.3 Status of the “information generates the universe” premise	26
15 Recommended Programs 61–70	28
16 Final judgment	29
17 Reproducibility	30
Archival note	31

0.1 Confidence convention

- **[Proven]** analytic theorem or exact finite algebra under the declared definitions.
- **[Strong evidence]** reproducible finite computation with a clear numerical margin.
- **[Moderate evidence]** robust only inside the explicitly tested finite class.
- **[Conditional]** depends on a declared regularizer, conversion scale, state, instrument, or sector axiom.
- **[Refuted]** contradicted by a proof, counterexample, or failed transfer test.

All core calculations in this monograph are dimensionless. The ontology that the nadsoliton is primordial information in a solitonic state is treated as the FIN starting premise, not as an experimentally established conclusion. No lower informational substrate is introduced.

This work exports no strict physical unit, no strict selector, no discharge of QW-2191, no completed legacy-to-strict bridge, no legacy-role transfer, no unit-bearing L_{total} , and no Theory-of-Everything closure.

Chapter 1

Executive summary

Programs 51–60 were designed after auditing:

- *Funkcja Greena w kontekście FIN — brakujące ogniwo*;
- *geometria przestrzeni stanów*;
- the current strict/legacy kernel split and provenance packets;
- the post-Programs-41–50 corrections.

The two source notes identify a valuable research direction: the Green resolvent can organize response, dual functional calculus, geometry, coarse graining, and adaptive information flow. Their central intuition survives, but several theorem claims do not.

The strongest corrected conclusion is:

The Green resolvent is a unifying response object, but it is not the missing physical principle.

It unifies finite mathematical descriptions. It does not determine an absolute unit, a preparation, an instrument, a physical clock, an environment, a detector, or the sign of an orientation sector.

The principal results are:

1. **The fixed value $s = 1.660307278766099$ belongs to strict, not legacy.** [Proven] It equals the strict C_{12} row sum to 4.4×10^{-16} . The canonical legacy row sum is -11.17405196 . Reusing strict s for legacy gives a minimum eigenvalue -8.54270057 , not a positive generator.
2. **Resolvent existence was stated too narrowly.** [Proven correction]

$$G(z) = (zI - A)^{-1}$$

exists exactly for $z \notin \sigma(A)$. Positivity is neither necessary for a nonreal resolvent nor sufficient for an unregularized graph-Laplacian inverse at $z = 0$, because a Laplacian has a zero mode.

3. **The proposed Green “resistance metric” theorem was overgeneralized.** [Refuted] For arbitrary positive-definite A ,

$$R_{ij} = G_{ii} + G_{jj} - 2 \operatorname{Re} G_{ij}$$

is a squared Hilbert distance. The guaranteed metric is $\sqrt{R_{ij}}$. An explicit three-state positive-definite counterexample gives a triangle excess 1.98 for R . Effective resistance R itself is a metric only because a connected graph Laplacian supplies additional structure.

4. **The same Green resolvent reconstructs both dynamics.** [Proven finite verification] Numerical Cauchy functional calculus reconstructs e^{-itA} , e^{-tA} , and A^{-1} with relative errors below 4×10^{-10} . This is a genuine unification of operator calculus, not a physical rule selecting an experimental record.

5. **A finite functional legacy-to-strict map exists on C_{12} but catastrophically fails transfer.** [Proven finite result] A degree-six map gives relative residual 9.25×10^{-14} on C_{12} . Applying the same map on C_{16} gives relative error 916.795. The C_{12} identity is finite spectral interpolation, not a continuum bridge or universality theorem.
6. **Exact fractal Green-Schur compression was constructed.** [Proven] Eliminating alternating states by a Schur complement preserves the Green response on retained states with residual 1.44×10^{-14} . This gives a rigorous information-compression map.
7. **That compression does not generate strict.** [Strong finite falsification] The canonical legacy-to-strict amplitude error is 0.99148. After one binary compression it is 0.95117, and after two it is 0.94673. Strict-phase tail fits have error approximately one; their fitted exponents are therefore non-identifiable and must not be interpreted as a derivation of $\eta = 1.8$.
8. **Fractal compression generates relative scale, not physical length.** [Proven obstruction] Binary elimination produces ratios 2^k and a dimensionless flow. It cannot choose ℓ_* , τ_* , or $\hbar_*$.
9. **Information loss under compression has a rigorous meaning.** [Proven conditional model] In the regularized Gaussian information model, Green-Schur compression reduces the legacy-versus-strict divergence from 23.63624 to 11.60157, consistent with data processing. This is dimensionless information loss, not automatically thermodynamic entropy or dissipated energy.
10. **The chiral receiver survives fractal compression but its sign remains unsourced.** [Proven] The compressed inversion-odd response has Frobenius norm 1.93502, while mirror covariance and oddness residuals are below 1.2×10^{-14} . The $+\lambda$ and $-\lambda$ branches remain isospectral. Compression transports chirality but does not select it.
11. **The shortest honest path to experimental physics has three layers.**

$$W_0 + \text{CA} + \text{OA} \quad \text{and, only for oriented sectors, SA.}$$

Here W_0 is the informational nadsoliton core, CA is dimensional calibration, OA is the operational map, and SA supplies a signed sector/origin when the observable requires it.

Chapter 2

Audit of the two source notes

2.1 What survives

The source notes correctly identify that a Green object:

- carries the complete spectral response of a finite self-adjoint generator;
- reconstructs unitary and diffusive functional calculi;
- supports perturbative resolvent identities;
- provides embeddings and graph-resistance geometries;
- is naturally compatible with state-dependent and adaptive operators;
- is the correct object for exact elimination by Schur complements.

This is a substantial mathematical unification.

2.2 Required corrections

Source claim	Corrected status
$A = sI - W$ is positive for strict and legacy because of a β threshold	Refuted. The quoted s is the strict row sum; canonical legacy gives an indefinite fixed- s operator
$G(z)$ exists only in a positive gapped phase $G(0) = A^{-1}$ for a graph Laplacian	Refuted. It exists whenever $z \notin \sigma(A)$ False without regularization or restriction. A connected Laplacian has one zero mode
$R_{ij} = G_{ii} + G_{jj} - 2G_{ij}$ is a metric for every $A > 0$	Refuted. $\sqrt{R}$ is the general Hilbert metric; R needs additional structure
The proof follows because R is a squared norm	Invalid. Squared norms do not generally satisfy the triangle inequality
A Dyson series is a renormalization theorem	Refuted. It is a resolvent identity; RG additionally requires a coarse-graining and normalization map
Same finite polynomial carrier implies the same universality class	Refuted. Universality is an asymptotic scaling statement, not finite interpolation
$\det G(z)$ is a topological invariant	Refuted as stated. It is a spectral invariant; topological meaning requires extra index/bundle structure
JSD between records measures only different contours	Refuted. It depends on preparation, dynamics, instrument, and record map

Source claim	Corrected status
$K(t) = e^{-tA}K(0)e^{-tA}$ is the Heisenberg picture	Refuted. Unitary Heisenberg evolution is $U_t^\dagger K U_t$; dissipative evolution requires a declared channel
A fitted d^η after repeated composition would prove a bridge	Insufficient. One needs a transferable coarse-graining law and a sourced fixed point

2.3 Corrected conceptual position

The Green resolvent is best understood as:

informational response + spectral memory + exact marginalization interface.

Calling it a propagator is mathematically natural only after specifying the operator, source, boundary conditions, and time/energy convention. Calling it a physical propagator requires additional CA and OA data.

Chapter 3

Minimal mathematical setup

On a finite cycle C_N , let

$$W_{ij} = K(d(i, j)), \quad L = \text{diag}(W\mathbf{1}) - W.$$

For positive weights, L is a graph Laplacian. Canonical legacy is signed, so its raw L is not automatically positive. When an inverse is required, this work declares a dimensionless regularized precision

$$A = L + \delta I, \quad \delta > -\lambda_{\min}(L).$$

The regularizer is a mathematical infrared control. It is not a physical mass.

The standard resolvent convention is

$$G(z) = (zI - A)^{-1}.$$

For functions analytic around the spectrum,

$$f(A) = \frac{1}{2\pi i} \oint_{\Gamma} f(z)(zI - A)^{-1} dz.$$

For a retained state set E and eliminated set O , write

$$A = \begin{pmatrix} A_{EE} & A_{EO} \\ A_{OE} & A_{OO} \end{pmatrix}.$$

The exact Green-Schur compressed operator is

$$A_{\text{eff}} = A_{EE} - A_{EO}A_{OO}^{-1}A_{OE},$$

and satisfies

$$A_{\text{eff}}^{-1} = (A^{-1})_{EE}.$$

This identity is the rigorous core of “fractal information compression” used in Programs 55, 57, and 58.

Chapter 4

Program 51 — Resolvent-domain and positivity audit

4.1 Question

Can the Green construction in the source note be applied unchanged to strict and canonical legacy?

4.2 Result

Quantity	Value
strict C_{12} row sum	1.6603072787660986
quoted s	1.660307278766099
canonical legacy row sum	−11.174051961650372
$\lambda_{\min}(sI - W_{\text{strict}})$	1.18×10^{-15}
$\lambda_{\min}(sI - W_{\text{legacy}})$	−8.5427005700
$\lambda_{\min}(L_{\text{legacy}, \text{signed}})$	−21.3770598104

The strict operator is a Laplacian to floating-point precision and therefore semidefinite, not positive definite. Canonical legacy requires a declared shift 21.6270598104 to place its signed Laplacian above the dimensionless floor 0.25.

4.3 Verdict

[Proven] The fixed- s formula is kernel-specific. The legacy Green inverse cannot inherit the strict shift. The Gram-PSD threshold of a different kernel matrix does not prove positivity of this generator.

Chapter 5

Program 52 — Corrected Green geometry

5.1 Question

Does the Green function automatically produce a metric information geometry?

5.2 Theorem

For $A > 0$, define $G = A^{-1}$ and

$$R_{ij} = G_{ii} + G_{jj} - 2 \operatorname{Re} G_{ij}.$$

There are vectors $v_i = G^{1/2} e_i$ such that

$$R_{ij} = \|v_i - v_j\|^2.$$

Therefore

$$d_G(i, j) = \sqrt{R_{ij}}$$

is a metric whenever distinct states have distinct embedded vectors.

The unsquared R is not a metric in general.

5.3 Explicit counterexample

The tested positive-definite Green matrix gives

$$R = \begin{pmatrix} 0 & 1.02 & 4.01 \\ 1.02 & 0 & 1.01 \\ 4.01 & 1.01 & 0 \end{pmatrix},$$

so

$$R_{13} - R_{12} - R_{23} = 1.98 > 0.$$

5.4 FIN finite results

- Strict shifted Green R happens to pass all triangles on C_{12} , but this is a finite property, not the general theorem.
- $\sqrt{R}$ passes by the Hilbert-embedding theorem.
- Effective resistance from the strict graph Laplacian is a metric.
- Both positive-part and absolute-value legacy repairs produce resistance metrics, but they produce different geometries.

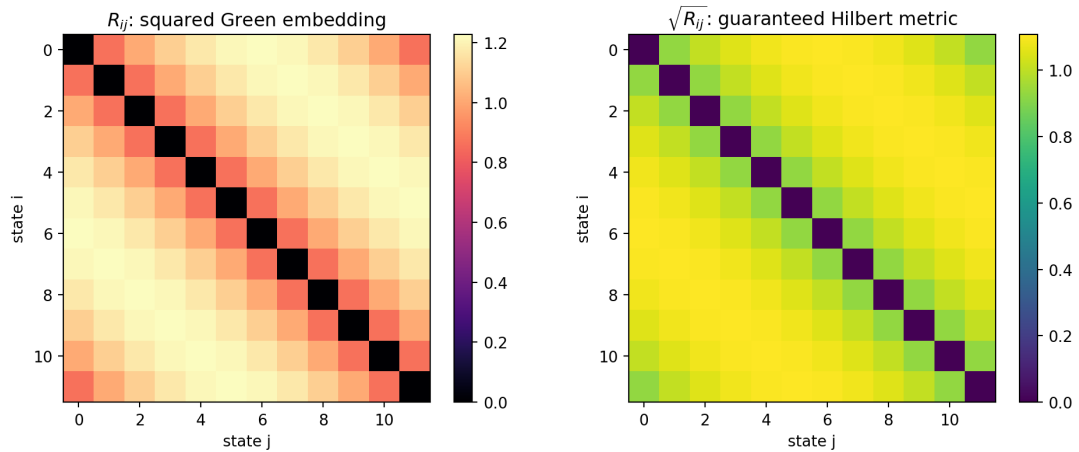


Figure 5.1: Squared Green embedding R and the guaranteed Hilbert metric $\sqrt{R}$ for the strict C_{12} realization.

5.5 Verdict

[Proven correction] The Green construction repairs the earlier $-\log |K|$ proposal only after choosing the correct distance and operator class. It does not define a unique physical spacetime metric.

Chapter 6

Program 53 — Resolvent reconstruction of dual dynamics

6.1 Question

Can one resolvent reconstruct both unitary and diffusive evolutions quantitatively?

6.2 Test

Using 1024 points on a Cauchy contour around the spectrum:

$$e^{-itA} = \frac{1}{2\pi i} \oint e^{-itz} (zI - A)^{-1} dz,$$

$$e^{-tA} = \frac{1}{2\pi i} \oint e^{-tz} (zI - A)^{-1} dz.$$

6.3 Results

Reconstruction	Relative Frobenius error
unitary e^{-itA}	1.65×10^{-10}
diffusive e^{-tA}	2.74×10^{-10}
static A^{-1}	3.65×10^{-10}
direct unitarity residual	9.66×10^{-16}

6.4 Verdict

[Proven finite verification] One spectral response object generates both calculi. However, an observed JSD or detector difference is not merely a contour difference. It depends on the full operational tuple:

(state, preparation, clock, dynamics, instrument, environment, apparatus, record).

Chapter 7

Program 54 — Functional Green bridge and transfer falsification

7.1 Question

Is the finite relation between legacy and strict a transferable Green-function bridge?

Because both C_{12} kernels are radial circulants, they share Fourier eigenspaces. A scalar functional relation exists exactly when every degeneracy class of W_{legacy} carries a single strict eigenvalue.

7.2 Results

Test	Value
distinct legacy eigenvalues on C_{12}	7
strict spread inside legacy degeneracy classes	1.11×10^{-16}
interpolating polynomial degree	6
C_{12} relative residual	9.25×10^{-14}
transfer error on C_{16}	916.795186

7.3 Interpretation

On C_{12} ,

$$W_{\text{strict}} = q_{12}(W_{\text{legacy}})$$

to numerical precision. Consequently their resolvents can also be related on this one finite spectrum.

But q_{12} is an interpolation through seven finite spectral points. Its failure on C_{16} shows that it is not support-independent.

7.4 Verdict

[**Refuted as a general bridge**] The finite functional identity does not establish:

- a continuum map;
 - a renormalization fixed point;
 - a universality class;
 - a source for strict phase, damping, or $\eta = 1.8$;
 - physical-role transfer.
-

Chapter 8

Program 55 — Green-Schur fractal compression

8.1 Question

Can an explicit, information-preserving fractal compression convert canonical legacy into strict?

8.2 New typed compression object

The tested map retains every second cycle state and exactly eliminates the remainder:

$$\mathcal{C}_2(A) = A_{EE} - A_{EO}A_{OO}^{-1}A_{OE}.$$

Repeated compression gives

$$\mathcal{C}_4 = \mathcal{C}_2 \circ \mathcal{C}_2.$$

Unlike an unspecified “coarse grain,” this map has an exact Green identity.

8.3 Results

Quantity	Value
24 → 12 retained-Green identity residual	1.44×10^{-14}
raw legacy→strict amplitude error	0.991483
one-step compressed error	0.951170
two-step compressed error	0.946726
one-step strict-phase tail-fit error	1.000000003
two-step strict-phase tail-fit error	1.000000000

The tail optimizations return extreme, incompatible exponents because the objective is essentially unable to fit the compressed profile. Those exponent values are not scientific estimates.

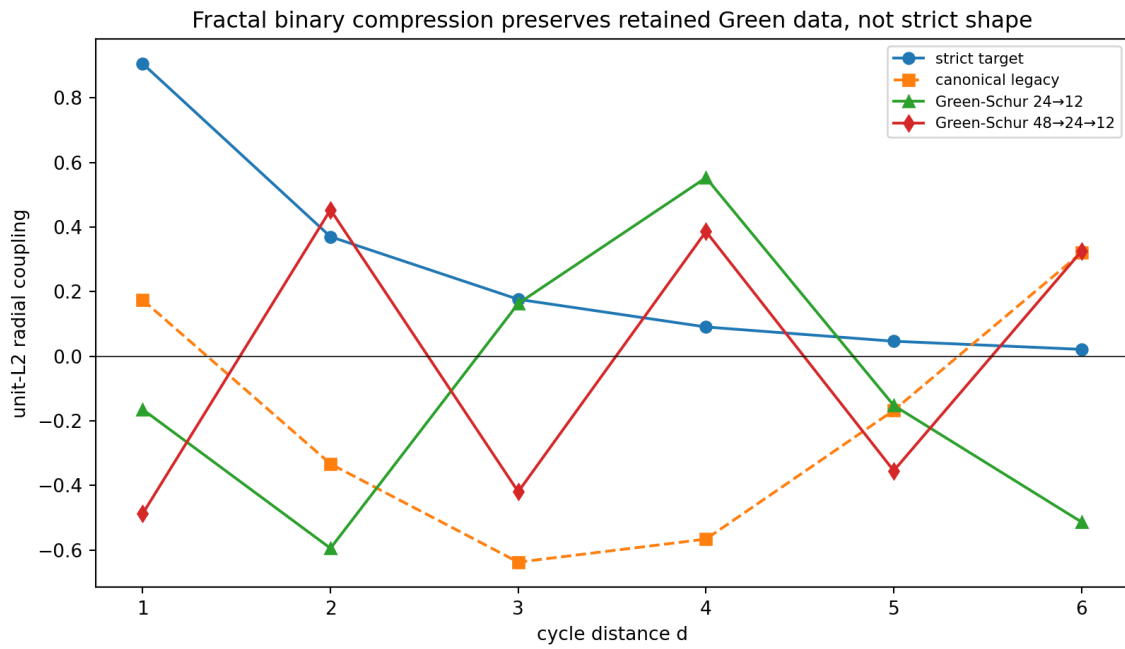


Figure 8.1: Exact binary Green-Schur compression preserves retained response but does not generate the strict profile.

8.4 Verdict

[Proven] The map is an exact marginalization of information response.

[Strong finite falsification] It does not generate the strict profile or $\eta = 1.8$.

[Proven obstruction] It supplies the relative scale ratios 2 and 4, but no absolute length.

This distinguishes three ideas often conflated under “fractal compression”:

1. exact elimination of hidden informational states — obtained;
2. flow toward the strict kernel — not obtained;
3. generation of physical scale — impossible without a scale-charged datum.

Chapter 9

Program 56 — Spectral dimension under repair and scale

9.1 Question

Does the heat trace recover a stable emergent dimension?

For nonzero Laplacian modes,

$$Z'(t) = \sum_{\lambda_k > 0} e^{-t\lambda_k}, \quad d_s(t) = -2 \frac{d \log Z'(t)}{d \log t}.$$

9.2 Results

Across raw strict C_{12} , strict C_{48} positive-part and absolute repairs, and both legacy repairs:

- the interval satisfying $d_s \in [0.85, 1.15]$ is only 0.120–0.137 decades wide;
- the apparent crossing time varies by almost two orders of magnitude;
- beyond the positive C_{12} strict support, even strict requires a declared repair because its long-distance extrapolation changes sign.

9.3 Verdict

[Refuted as a dimension export] There is no robust spectral-dimension plateau in the tested finite family. A curve crossing $d_s = 1$ is not evidence that the theory has generated one-dimensional physical space.

[Moderate evidence] Heat-trace diagnostics remain useful for comparing declared operator realizations.

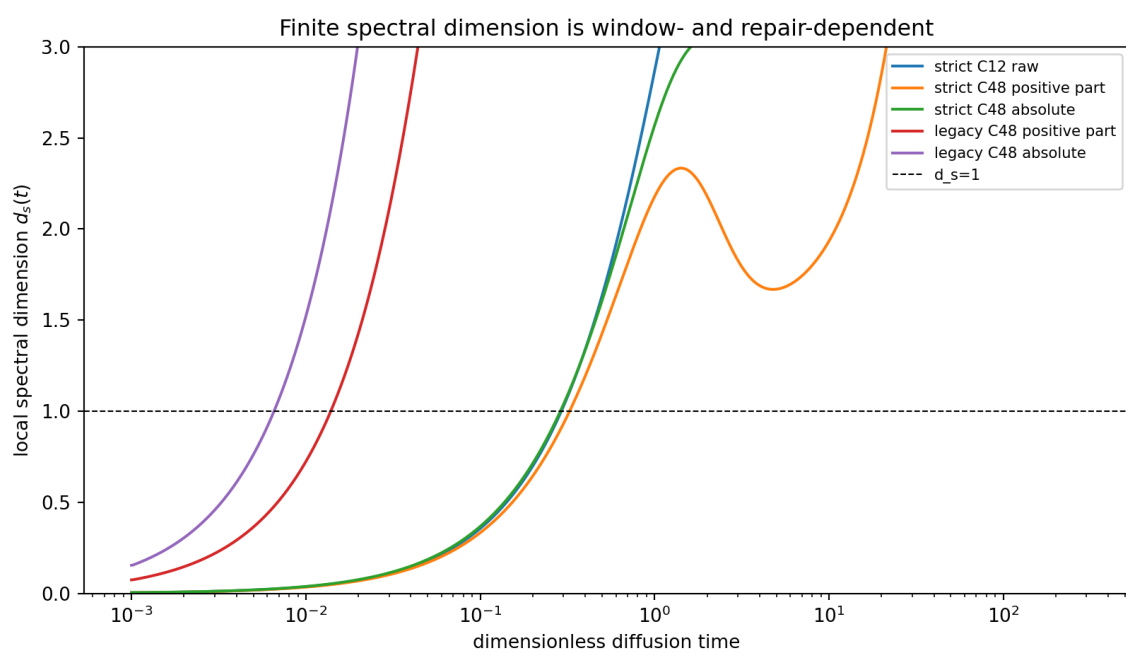


Figure 9.1: Finite local spectral dimension depends on support, diffusion window, and positivity repair.

Chapter 10

Program 57 — Information action and compression loss

10.1 Question

Can Green compression give a rigorous form of information loss?

Treat a regularized $A > 0$ as a Gaussian precision and $G = A^{-1}$ as its covariance. The Schur complement is then the exact marginal precision on retained informational states.

10.2 Results

Quantity	Value
full $D_{\text{KL}}(\text{legacy} \text{strict})$	23.63624087
retained-state D_{KL}	11.60157427
data-processing margin	12.03466660
legacy marginal-precision identity residual	2.56×10^{-13}
strict identity residual	8.95×10^{-16}

10.3 Verdict

[Proven within the Gaussian information model] Eliminating states reduces model distinguishability and transfers their influence into an effective operator.

This is the mathematically correct sense in which fractal compression can create apparent information loss while the larger description retains the full covariance.

It is not yet:

- thermodynamic entropy;
- heat;
- Landauer work;
- physical action;
- an experimentally calibrated energy.

Those identifications require temperature, k_B , an environment protocol, a clock, and a work/heat instrument.

Chapter 11

Program 58 — Chiral Green response under fractal compression

11.1 Question

Does an inversion-odd mirror coupling disappear under coarse graining, generate its own sign, or survive as an effective sector?

Let

$$C = \frac{i}{2}(D - D^T), \quad RCR = -C, \quad A_\lambda = A_0 + \lambda C.$$

11.2 Results

Test	Value
full odd-carrier residual	0
full branch covariance residual	1.42×10^{-14}
compressed branch covariance residual	1.14×10^{-14}
compressed odd-response residual	3.63×10^{-16}
compressed chiral response norm	1.93501688
compressed $+\lambda/-\lambda$ spectral residual	0

11.3 Verdict

[Proven] Green-Schur compression transports the correct chiral representation object.

[Refuted] It does not select the sign. Reflection still exchanges two isospectral branches:

$$RA_{+\lambda}R = A_{-\lambda}.$$

Fractal compression can preserve a sector, amplify or suppress its response, and integrate out hidden states. It cannot source the sector's orientation from a radial scalar kernel.

Chapter 12

Program 59 — Operational records from one generator

12.1 Question

What is the smallest experimentally shaped calculation that can be made without claiming a physical calibration?

A coherent two-source preparation was evolved with the strict C_{12} generator by:

$$U_t = e^{-itL}, \quad T_t = e^{-tL}.$$

For the unitary branch the record uses Born populations. For the diffusive branch it uses the stochastic population vector.

12.2 Results

Quantity	Value
unitary probability sum	$1 - 1.1 \times 10^{-16}$
diffusive probability sum	$1 - 2.2 \times 10^{-16}$
minimum heat-kernel entry	0.0256684
coherent-incoherent L^1 difference	0.182669
wave-diffusion JSD, fine record	0.0428693
wave-diffusion JSD after pair coarse graining	0.0352858
JSD data-processing margin	0.00758347

12.3 Verdict

[Proven finite operational statement] One generator can support distinct wave and diffusion records once the preparation and instrument are specified. Coarse graining reduces their distinguishability.

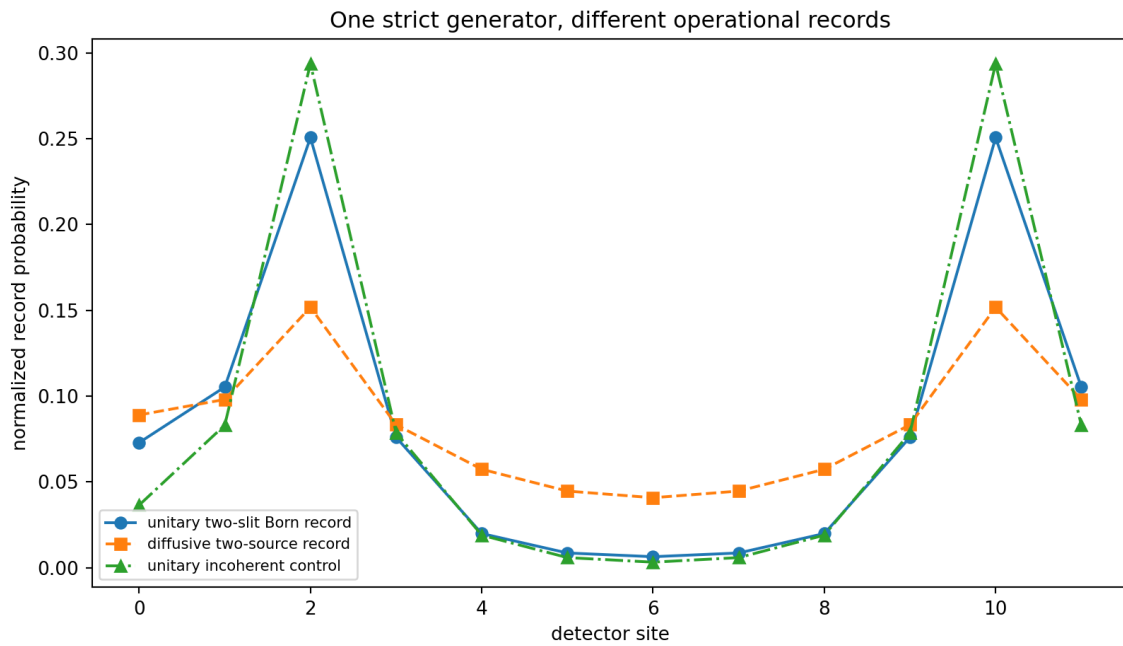


Figure 12.1: Unitary coherent, diffusive, and incoherent-control records from one declared strict generator.

The observer paradox is therefore operationally avoidable. The missing objects are not consciousness but:

- state and preparation;
- clock;
- chosen dynamical calculus;
- instrument;
- environment;
- apparatus;
- record map;
- dimensional calibration.

Chapter 13

Program 60 — Physical-scale obstruction and minimal conditioned bridge

13.1 Question

Can information, the Green response, or fractal compression determine dimensional physics without imported calibration?

Let the strict dimensionless spectral gap be

$$\hat{\Delta} = 0.7541211542070796.$$

After a conversion triple $(\ell_*, \tau_*, \hbar_*)$,

$$E_{\text{gap}} = \frac{\hbar_*}{\tau_*} \hat{\Delta}, \quad c_* = \frac{\ell_*}{\tau_*}.$$

Three admissible conversion gauges leave every dimensionless FIN computation unchanged but generate energy gaps ranging from

$$7.54 \times 10^{-6} \quad \text{to} \quad 7.54 \times 10^8$$

in their declared unit coordinates.

13.2 Impossibility theorem

Suppose all W_0 outputs are dimensionless and invariant under independent rescalings of length, time, and action units. Then no function of W_0 alone can select a unique nonzero value carrying any of those dimensions.

The proof is the positive-torsor argument: an invariant input cannot equivariantly select a representative of a freely rescaled unit torsor.

Fractal compression does not evade the theorem. It gives:

$$d \mapsto 2^k d,$$

which is a relative scale transformation. The physical statement

$$x = 2^k \ell_* d$$

still contains the unsourced ℓ_* .

13.3 Minimal conditioned bridge

The shortest honest architecture is:

13.3.1 W_0 : informational core

- informational nadsoliton;
- canonical strict and canonical legacy kernels with their distinct roles;
- Green response;
- spectral calculus;
- exact Schur compression;
- dimensionless information geometry.

13.3.2 CA: conversion axioms

A basis equivalent to:

$$(\ell_*, \tau_*, \hbar_*).$$

Relations such as a calibrated limiting speed may reduce the number of independently reported constants, but the relation itself is additional physical input.

13.3.3 OA: operational axioms

$$(\rho_0, \mathcal{P}, \mathcal{T}, \mathcal{I}, \mathcal{E}, \mathcal{A}, \mathcal{R}),$$

representing state, preparation, clock/dynamics, instrument, environment, apparatus, and record.

13.3.4 SA: sector axioms, only when required

$$(r_0, \lambda_0, \text{coupling law})$$

for origin-, polarity-, or chirality-sensitive records.

13.3.5 W_3 : conditioned physics

Only after CA and OA can a dimensionless spectrum be mapped to measured time, energy, length, and probability records. SA is additionally required only when the experiment distinguishes mirror sectors.

13.4 Verdict

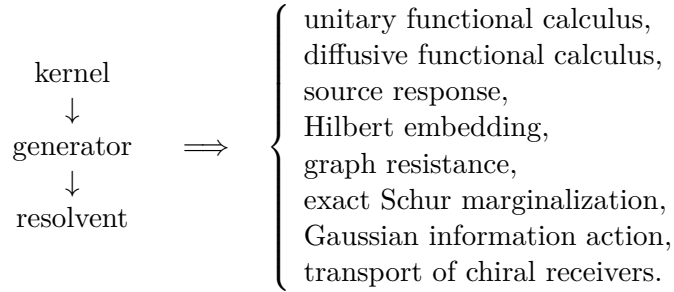
[**Proven obstruction**] Information and fractal ratios do not determine dimensional calibration.

[**Conditional bridge**] $W_0 + CA + OA(+SA)$ is sufficient to define an experimentally shaped model, but its predictions are conditioned on those additional axioms until they are independently sourced.

Chapter 14

Integrated scientific conclusions

14.1 What the Green function genuinely unifies



This is a real mathematical synthesis.

14.2 What it does not unify

The resolvent does not by itself provide:

- a unique legacy-to-strict completion;
- a stable continuum universality class;
- $\eta = 1.8$;
- an absolute length or time;
- $\hbar$, energy, or mass;
- a physical boundary condition;
- a state preparation;
- Born's rule;
- an instrument or apparatus;
- an environment;
- a signed selector.

14.3 Status of the “information generates the universe” premise

The premise is mathematically productive when interpreted as:

physical sectors are emergent operational representations of one informational nadsoliton, not fields placed on a deeper independent informational medium.

Programs 51–60 show how information can generate:

- relations;
- response;
- distinguishability;
- effective geometry;
- relative scale;
- coarse-grained dynamics;
- apparent information loss;
- sector receivers.

They do not show how dimensionless information internally generates the calibration that identifies one such representation with measured metres, seconds, joules, particles, or laboratory apparatus.

The missing bridge is therefore not “more information.” It is a mathematically typed map from informational relations to operationally calibrated records.

Chapter 15

Recommended Programs 61–70

The next research stage should avoid repeating finite interpolation or unsourced scalar fitting.

1. **Program 61 — Continuum functor for Green-Schur compression.**
Define a projective family $C_{2^k N} \rightarrow C_N$ and test resolvent convergence under one fixed normalization.
 2. **Program 62 — Regularizer-independence audit.**
Determine which Green, KL, and compression conclusions survive $\delta \rightarrow 0^+$, quotienting the zero mode, and pseudoinverse replacement.
 3. **Program 63 — Compression semigroup theorem.**
Test whether the Schur maps compose associatively up to canonical relabeling and characterize their fixed points.
 4. **Program 64 — Signed-kernel Krein-space alternative.**
Compare positive regularization against an indefinite inner-product formulation without silently calling either one physical.
 5. **Program 65 — Operational tomography of the dual calculus.**
Determine the minimum preparations and instruments required to distinguish U_t from T_t after coarse graining.
 6. **Program 66 — Fractal causal-order candidate.**
Add a declared partial order and test whether Green support respects finite propagation or produces unavoidable instantaneous tails.
 7. **Program 67 — Physical calibration identifiability.**
Given synthetic clock, length, and energy records, determine the minimum number of independent experiments needed to identify CA.
 8. **Program 68 — Chiral source-law falsification.**
Test one explicit nadsoliton-to- λC formula, including origin covariance and sign provenance, rather than enumerating receiver classes.
 9. **Program 69 — Information-to-thermodynamics protocol.**
Add an explicit bath, temperature, erasure channel, and work instrument; test Landauer inequalities without identifying Shannon entropy with heat by definition.
 10. **Program 70 — Blinded physical-model challenge.**
Generate records from a hidden conditioned model and compare strict, legacy, compressed, and null models without including the exact generator as an announced target.
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Chapter 16

Final judgment

The most important new object is not merely $G = A^{-1}$. It is the exact three-part structure:

Green response + fractal Schur compression + operational calibration.

The first two are now mathematically executable inside FIN. The third is the minimal missing bridge to our physics.

The corrected deepest conclusion is:

The informational nadsoliton can consistently generate a dimensionless relational world with response, geometry, dual dynamics, coarse-grained information loss, and chiral sector receivers. Fractal compression organizes relative scale and effective laws, but it cannot generate an absolute physical unit or choose an orientation branch. Experimentally testable physics begins only after a calibrated operational map is supplied.

Chapter 17

Reproducibility

- Executable: `fin_programs_51_60_green_fractal_physics.py`
- Regression tests: `test_fin_programs_51_60_green_fractal_physics.py`
- Machine-readable results: `FIN_Programs_51_60_Green_Fractal_Physics_Results.json`
- Figures: `FIN_Programs_51_60_Green_Fractal_Physics_Figures/`

Suggested citation:

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Archival note

Machine-readable results are stored in `FIN_Programs_51_60_Green_Fractal_Physics_Results.json`. The executable and ten regression tests reproduce the finite results and all four figures.

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